

Chemistry A

Twenty First Century Science Suite

OCR GCSE in Chemistry A J634

Foreword to the Second Edition

This Second Edition of the OCR GCSE Chemistry A specification has been produced to correct minor errors found in the original edition (published in Dec 2005). There are no changes to actual content or the scheme of assessment. Centres should note however the grade descriptions in Appendix A have now been replaced with the correct versions.

Section 6.6 has been updated (amended in Oct 2007).

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1.1 About the Twenty First Century Science Suite

The Twenty First Century Science suite comprises six specifications which share common material, use a similar style of examination questions and have a common approach to skills assessment. The qualifications available as part of this suite are:

GCSE Science A (J630)	which emphasises scientific literacy – the knowledge and understanding which candidates need to engage, as informed citizens, with science-based issues. As with other courses in the suite, this qualification uses contemporary, relevant contexts of interest to candidates, which can be approached through a range of teaching and learning activities.
GCSE Additional Science A (J631)	which is a concept-led course developed to meet the needs of candidates seeking a deeper understanding of basic scientific ideas. The course focuses on scientific explanations and models, and gives candidates an insight into how scientists develop our scientific understanding of ourselves and the world we inhabit.
GCSE Additional Applied Science A (J632)	which meets the needs of candidates who wish to develop their scientific understanding through authentic, work related contexts. The course focuses on procedural and technical knowledge that underpins the work of practitioners of science and gives candidates an insight into what is involved in being a practitioner of science.
GCSE Biology A (J633)	each of which provides an opportunity for further developing an understanding of science explanations, how science works and the study of elements of applied science, with particular relevance to professional scientists.
GCSE Chemistry A (J634)	
GCSE Physics A (J635)	

This suite is supported by the Nuffield Curriculum Centre and The University of York Science Education Group.

1.2 About this Chemistry Specification

This booklet contains OCR's GCSE specification in Chemistry for teaching from September 2006 and first certification in June 2008.

This specification aims to provide candidates with the scientific understanding needed to progress to further studies of chemistry, should they choose to undertake them. Candidates should gain an insight into:

- what is involved in being a practitioner of science;
- how scientists develop scientific understanding of ourselves and the world we inhabit;

ates also need to be able to reflect on scientific knowledge itself, the practices that have led to it, the kinds of reasoning that are used in developing a scientific argument, and on the implications that arise when scientific knowledge is put to practical use. These are referred to as 'Ideas in Science' (IaS). This specification provides a combination of these two essential elements.

The specification comprises seven teaching modules which are assessed through four units. Candidates take Units 1, 2 and 3 **and** either Unit 4 **or** 5.

Unit Code	Title	Duration	Weighting	Total Mark
A321	Chemistry A Unit 1 – modules C1, C2, C3	40 mins	16.7%	42
A322	Chemistry A Unit 2 – modules C4, C5, C6	40 mins	16.7%	42
A323	Chemistry A Unit 3 – Ideas in Context plus C7	60 mins	33.3%	55
A329	Chemistry A Unit 4 – Practical Data Analysis and Case Study	-	33.3%	40
A330	Chemistry A Unit 5 – Practical Investigation	-	33.3%	40

Qualification Titles and Levels

The qualification is shown on a certificate as OCR GCSE in Chemistry.

The qualification is approved by the regulatory authority, QCA, as part of the National Qualifications Framework.

Candidates who gain grades G to D will have achieved an award at Foundation Level (Level 1 of the National Qualifications Framework).

Candidates who gain grades C to A* will have achieved an award at Intermediate Level (Level 2 of the National Qualifications Framework).

1.4 Aims

The aims of this GCSE specification are to encourage candidates to:

- acquire a systematic body of scientific knowledge, and the skills needed to apply this in new and changing situations in a range of domestic, industrial and environmental contexts;
- acquire an understanding of scientific ideas, how they develop, the factors which may affect their development and their power and limitations;
- plan and carry out investigative tasks, considering and evaluating critically their own data and that obtained from other sources, and using ICT where appropriate;
- use electronic (internet, CD ROMs, databases, simulations etc.) and/or more traditional sources or information (books, magazines, leaflets etc.) to research and plan an investigation;
- select, organise and present information clearly and logically, using appropriate scientific terms and conventions, and using ICT where appropriate;
- interpret and evaluate scientific data from a variety of sources.

1.5 Prior Learning/Attainment

Candidates who are taking courses leading to this qualification at Key Stage 4 should normally have followed the corresponding Key Stage 3 programme of study within the National Curriculum.

Other candidates entering this course should have achieved a general educational level equivalent to National Curriculum Level 3.

–6 are designed to be taught in approximately half a term, in 10% of the candidates' curriculum time. Module C7 is designed to be taught in only one and a half terms at 10% curriculum time.

defines the required teaching and learning outcomes.

ation content in is displayed as seven modules. The titles of these seven modules are listed below.

Air Quality	Module C2: Material Choices	Module C3: Food Matters
chemicals make up air, and which ones are How do I make sense of data about air chemical reactions produce air pollutants? What these pollutants in the atmosphere? ion harmful to me, or to my environment? es can we make personally, locally, nationally to improve air quality?	• What different properties do different materials have? • Why is crude oil important as a source of new materials such as plastics and fibres? • Why does it help to know about the molecular structure of materials such as plastics and fibres? • When buying a product, what else should we consider besides its cost and how well it does its job? How should we manage the wastes that arise from our use of materials?	• What is the difference between intensive and organic farming? • Why are chemicals deliberately added to food? • How can we make sure that our food does not contain chemicals that may be harmful to health? • Why does what we eat affect our health?
Chemical Patterns	Module C5: Chemicals of the Natural Environment	Module C6: Chemical Synthesis
he patterns in the properties of elements? emists explain the patterns in the properties of its? emists explain the properties of compounds of nd Group 7 elements?	• What types of chemicals make up the atmosphere and hydrosphere? • What types of chemicals make up the Earth's lithosphere? • Which chemicals make up the biosphere? • How can we extract useful metals from minerals?	• Chemicals and why we need them • Planning, carrying out and controlling chemical synthesis
Further Chemistry		
carboxylic acids and esters anges in chemistry reactions and equilibria mistry		

Module Content

Content of modules C1, C2 and C3 is based on a set of Science Explanations about Science (see Appendices F and G). The presentation of the content of these modules reflects these ideas about science in the presentation of the synopsis page, which has the following structure:

	Questions that science may help to answer
Key sense of data about air	e.g. What chemicals make up air, and which ones are pollutants?
Concepts	Ideas about Science
Examples	e.g. IaS 1 Data and its limitations

Identifies:

Concepts which are likely to be uppermost in the minds of citizens when considering air quality, whatever their understanding of science;

Issues about the topic that science can help to address which could reasonably be asked of a literate person;

Explanations and Ideas about Science which are introduced or further developed in the module.

Modules C6 and C7 also begin with a synopsis page, which outlines the content of the module.

Additional content is provided to give teachers additional information, expressed in terms of the way in which the content is linked to other parts of the specification, and to summarise this information.

Explanation and guidance

These content statements will only be assessed on Higher Tier papers.

Advisory notes for teachers to clarify depth of cover required.

The quality of air is becoming a major world concern. In this module, candidates explore environmental and health consequences of certain air pollutants, and options for improving air quality in the future. The emphasis is on health issues arising from burning fuels, rather than global issues such as climate change.

Candidates learn about the chemical relationship between the burning of fossil fuels and the production of air pollutants. This module introduces molecular elements and compounds to illustrate chemical explanations.

By analysing their own and given data on concentrations of pollutants, candidates learn about the way in which scientists use data, and also that all data have certain limitations.

Issues for citizens	Questions that science may help to answer
How do I make sense of data about air pollution?	What chemicals make up air, and which ones are pollutants?
Where do pollutants come from?	What chemical reactions produce air pollutants?
Is air pollution harmful to me or my environment?	What happens to pollutants in the atmosphere?
How can we improve air quality?	What choices can we make personally, locally, nationally or globally to improve air quality?
Science Explanations	Ideas about Science
SE1 Chemicals	laS 1 Data and its limitations
SE2 Chemical change	laS 2.1, 2.3–2.5 Correlation and cause
	laS 4.2 The scientific community
	laS 6.3 Making decisions about science and technology

ICT Opportunities

This module offers opportunities for illustrating the use of ICT in science, for example:

- collecting, storing and displaying data from a large network of measuring instruments;
- displaying data in a variety of charts, graphs and maps for analysis and evaluation.

Use of ICT in teaching and learning can include:

- internet to research local air quality data;
- animation to illustrate chemical change during reactions;
- simulation to model effects of local government policy decisions on air quality;

1. recall that the Earth is surrounded by an atmosphere made up mainly of nitrogen, oxygen and argon, plus small amounts of water vapour, carbon dioxide, and other gases;
2. recall that the relative proportions of gases in the atmosphere are about 78% nitrogen, 21% oxygen and 1% argon;
3. recall that human activity adds small amounts of carbon monoxide, nitrogen oxides and sulfur dioxide to the atmosphere;
4. recall that human activity also adds extra carbon dioxide and small particles of solids (e.g. carbon) to the atmosphere;
5. recall that some of these substances, called pollutants, are directly harmful to humans and some are harmful to the environment and so cause harm to humans indirectly;
6. when using their own and given data relating to measured concentrations of atmospheric pollutants, or the composition of the atmosphere:
 - uses data rather than opinion in justifying an explanation;
 - can suggest reasons why a measurement may be inaccurate;
 - can suggest reasons why several measurements of the same quantity may give different results;
 - when asked to evaluate data, makes reference to its reliability (i.e. is it repeatable?);
 - can calculate the mean of a set of repeated measurements;
 - from a set of repeated measurements of a quantity, uses the mean as the best estimate of the true value;
 - can explain why repeating measurements leads to a better estimate of the quantity;
 - can make a sensible suggestion about the range within which the true value of a measured quantity probably lies;
 - **can justify the claim that there is/is not a 'real difference' between two measurements of the same quantity;**
 - can identify any outliers in a set of data, and give reasons for including or discarding them.